BidPace-RL: Budget-Constrained Real-Time Ad Bidding

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Abstract. We study hourly bidding under a hard daily budget using chronological replay of iPinYou auction logs. Original NumPy implementations compare DQN, two PPO controllers and an empirical model-based RL candidate with ten deployable baselines. On five held-out advertiser-days, the empirical candidate obtains 50.2 recorded clicks per day, versus 45.2 for the highest-scoring baseline, an observed 11.06% gain. However, validation selects DQN, which obtains only 41.6. The empirical candidate also leads baseline means in six declared stress scenarios. Its result remains a secondary, exploratory finding: only three dates and two campaigns are evaluated, and intervals against PID and traffic-linear include zero. All 1,225 submitted evaluation rows respect the budget.

1 Problem and data

An advertiser must allocate a finite budget across a day of auctions. Spending early can prevent later clicks, while overly conservative bids leave useful budget unspent. We model hourly bidding as an approximate budget-control MDP with a 24-step horizon. The policy observes 13 normalized budget, time, quality and completed-hour feedback features and chooses a multiplier $\alpha_h \in [0,3]$. Bids scale with training-fitted hourly CTR. Reward is $R_h = \sum_{i \in \mathscr{W}_h} y_i$, where $\mathscr{W}_h$ contains affordable winning impressions and $y_i \in \{0,1\}$ is a recorded click. The daily constraint is $\sum_h c_h \le B$.

The selected iPinYou input [1] contains 15,337,988 priced impressions: public Season 2 training logs from June 6–12 and Season 3 logs from October 19–27, 2013. Nine campaigns have training profiles. October 25–27 supplies five observed advertiser-days: campaigns 2261 and 2997 on the first two dates and 2261 on the last. Missing campaign-days are not invented. Calibration uses training prefixes only; budget is 30% of a campaign's mean training cost over its active days.

The simulator preserves timestamp order, pays the logged price when a bid wins, skips unaffordable impressions, and permits affordable later arrivals. It stops at day end or budget exhaustion. Delayed click labels are masked at each fitting cutoff. Lost-auction prices and counterfactual responses are unavailable, so the estimand is *conditional replay of logged winning inventory*, not live whole-market performance.

2 Learning and control

Neural agents. DQN uses a 48/32 ReLU network, ten discrete multipliers, three-step replay, Double-DQN targets [2] and a target network. PPO [3] uses a Gaussian latent actor, a value critic, clipped ratios and full-episode advantages with $\gamma = \lambda = 1$. Traffic PPO learns relative bid changes; residual PPO learns bounded multiplicative corrections around a causal cost controller. Their 44 inputs include causal history. Neural policies share parameters across campaigns and omit advertiser IDs from their observations.

Empirical model-based RL. Bootstrap samples of active training days estimate hourly reward $\hat{r}_h(a)$ and normalized cost $\hat{c}_h(a)$. Backward Bellman backups learn Q-values on 401 budget states, with interpolated continuation values:

$$Q_h(b,a) = f(b,\hat{c}_h(a))\hat{r}_h(a) + \max_{a'} Q_{h+1}(b - \min\{b,\hat{c}_h(a)\}, a'), \qquad (1)$$

where $f(b,c) = \min(1, b/c)$ for $c > 0$, and $f(b,0) = 1$. Exhausted-budget and terminal values are zero. Proportional partial-hour fulfillment is a model approximation; evaluation always uses exact arrival-order affordability.

Actions combine the DQN grid with 25 logarithmically spaced multipliers. Reward estimates mix recorded training clicks and smoothed CTR times eligible inventory. Validation selects weight 0.5 from $\{0, 0.5, 1\}$. Unobserved hours use a pooled training-hour prior. Execution adapts to past price and inventory feedback. This learner uses campaign-specific tables, so comparison with a shared neural agent changes representation as well as algorithm. It is model-based value iteration, not conventional tabular Q-learning.

3 Experimental protocol

Forward folds validate October 22, 23 and 24, each using only earlier dates plus the June auxiliary data. Each fold refits CTR, traffic, price and budget calibration. Validation averages six conditions: budgets 0.20/0.30/0.45, and budget 0.30 at prices $\times 1.20$ and coverage 0.50/0.75. The selection objective is mean paired gain over the strongest tuned validation baseline minus 0.5 times its across-date standard deviation.

Seeds are 11, 23, 37, 53 and 71. Ensembles average Q-values or latent actor means *before acting*; their scores are not averages of member scores. Neural training randomizes budgets in $[0.18, 0.50]$, prices in $[0.85, 1.25]$ and retained coverage in $[0.35, 1]$. Coverage preserves price/response pairing. These transformations are augmentation, not additional historical data.

Comparators include constant, global/campaign/budget linear, PID, quality PID, market/traffic-adaptive linear, unscaled linear and a cost controller. All share the bid cap of 3. A privileged linear method is a separate diagnostic. The primary selection is frozen before evaluation. DQN is selected at 1,500 episodes; both PPO variants use 4,096. Its development objective is -5.43, versus -17.96 for the selected empirical mixture.

4 Results

Table 1 reports all nominal methods. The primary DQN yields 41.6 clicks/day, below traffic-adaptive linear's 42.0 and unscaled linear's 45.2. The secondary empirical candidate yields 251 clicks over five advertiser-days (50.2/day), versus 226 (45.2/day) for unscaled linear. Its observed lift is 11.06%; aggregate spend per click is 14.22% lower. Mean daily utilization is 87.87%. Member scores span 48.2–50.8 clicks/day.

Table 1: Nominal held-out means over five advertiser-days. Utilization averages daily spend/budget fractions. P: validation-selected primary; S: secondary candidate; D: privileged diagnostic.

Method	Clicks/day	Util. (%)
Constant	41.6	100.00
Global linear	42.0	100.00
Campaign linear	43.2	92.33
Budget linear	43.2	92.33
PID	45.0	99.31
Quality PID	45.0	99.36
Market linear	43.4	93.85
Traffic linear	42.0	97.24
Unscaled linear	45.2	86.37
Cost controller	39.6	71.04
Traffic DQN (P)	41.6	87.88
Traffic PPO	42.0	100.00
Residual PPO	40.2	66.89
Empirical RL (S)	50.2	87.87
Privileged linear (D)	41.4	100.00

Table 2: Empirical RL versus the largest deployable baseline mean in each scenario. The baseline is chosen descriptively after evaluation, not as a deployment selector.

Scenario	Emp. RL	Best baseline	Lift (%)
Nominal	50.2	Unscaled: 45.2	+11.06
Prices $\times 1.5$	39.6	Q-PID: 37.0	+7.03
Evening $\times 3$	57.6	Unscaled: 55.4	+3.97
Budget 0.15	30.8	Market: 28.0	+10.00
Budget 0.60	75.8	Constant: 71.2	+6.46
Quality 0.5	27.4	Unscaled: 25.0	+9.60
Coverage 0.5	40.6	Constant: 38.6	+5.18

The empirical candidate exceeds baseline means in all seven declared conditions (Table 2). The six stresses reuse the same five advertiser-days. Evening multiplication duplicates opportunities; quality and coverage stresses thin labels or inventory with paired seeds. These rows do not constitute seven independent real-world datasets.

Benefits differ by campaign. Empirical RL averages 98.5 clicks/day on 2997, versus unscaled linear's 81.0; on 2261 it averages 18.0 versus 21.33. Figure 1 shows the paired heterogeneity. Thus the aggregate gain does not imply a win on every advertiser-day or campaign.

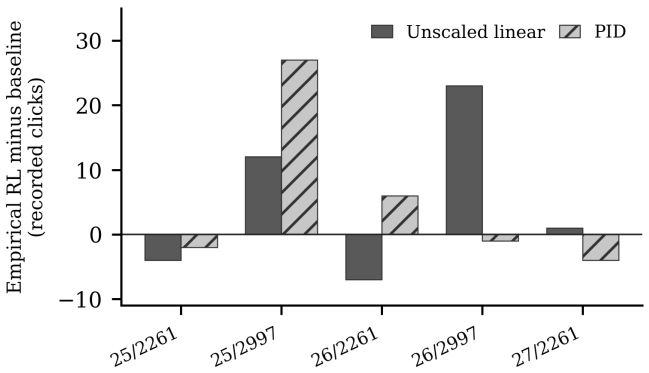


Figure 1: Paired nominal differences on five advertiser-days in October 2013. Tick labels give date/campaign. Both positive and negative differences are retained.

5 Uncertainty, checks and conclusion

Exact date-cluster bootstraps enumerate all $3^3 = 27$ ordered date resamples and retain unequal campaign-day counts. Unadjusted 95% intervals for empirical-minus-baseline clicks/day are $[-2.0, 14.5]$ against traffic-linear, $[-4.0, 12.5]$ against PID, and $[1.0, 8.0]$ against unscaled linear. Three date clusters and multiple secondary comparisons limit inferential strength; five optimization/bootstrap seeds are not independent test dates.

The uploaded evaluation matches its frozen SHA-256; code fingerprints match the source release. Audits reproduce all 280 submitted differences/intervals and check the complete 1,225-row grid, finite values, budget compliance and click-aware bounds. Source tests cover gradients, causal cutoffs, affordability and the full CLI. Checkpoint-level replay was not repeated from the submitted metric files.

The project demonstrates a promising empirical model-based controller under hard budgets, but also a failure of validation-based family selection: DQN remains the primary negative result. The empirical gain is a pre-specified secondary finding, not proof of general superiority. Confirmation requires a fixed choice evaluated on additional untouched dates and campaigns. Sparse campaign histories, approximate transitions and conditional inventory replay remain material limitations.

References

[1] W. Zhang, S. Yuan, J. Wang, and X. Shen. *Real-Time Bidding Benchmarking with iPinYou Dataset*. arXiv:1407.7073, 2014 (rev. 2015).

[2] H. van Hasselt, A. Guez, and D. Silver. *Deep Reinforcement Learning with Double Q-learning*. AAAI, 2016. arXiv:1509.06461.

[3] J. Schulman et al. *Proximal Policy Optimization Algorithms*. arXiv:1707.06347, 2017.